

[< Back](#)

Advertis

Clinical Implant Dentistry and Related Research / Volume 17, Issue S1 / pp. e1-e16

REVIEW

Immediate Placement of Implants into Infected Sites: A Systematic Review

Bruno Ramos Chrcanovic DDS, MSc , Maximiliano Delany Martins PhD,
Ann Wennerberg DDS, PhD

First published: 02 July 2013

<https://doi.org/10.1111/cid.12098>

 [VIEW METRICS](#)

Accessibility issue? [Request accessibility update.](#)

Conflict of interest statement

Bruno Ramos Chrcanovic: No conflict of interest

Maximiliano Delany Martins: No conflict of interest

Ann Wennerberg: No conflict of interest



Abstract

Background

Traditionally, before placing dental implants, the compromised teeth are removed and the extraction sockets are left to heal for several months. To preserve the alveolar bone level from the collapse caused by healing and to reduce treatment time in situations in which tooth extraction precedes implant placement, some clinicians began to install the implant immediately into the postextraction socket without waiting for the site to heal.

[< Back](#)

The purpose of this study was to review the literature regarding treatment outcomes of immediate implant placement into sites exhibiting pathology after clinical procedures to perform the decontamination of the implant's site. The following questions were raised: Does the presence of periodontal or endodontic infection affect immediate implant placement success? What is suggested to address the infection in the socket prior to immediate placement?

Materials and Methods

An electronic search in PubMed (U.S. National Library of Medicine, Bethesda, MD, USA) was undertaken in March 2013. The titles and abstracts from these results were read to identify studies within the selection criteria. Eligibility criteria included both animal and human studies, and excluded any review and case reports articles. The publication's intervention had to have been implant placement into a site classified as having an infection (periapical, endodontic, perioendodontic, and periodontal).

Results

The search strategy initially yielded 706 references. Thirty-two studies were identified within the selection criteria, from which nine were case reports and review articles and were excluded. Additional hand-searching of the reference lists of selected studies yielded five additional papers.

Conclusions

The high survival rate obtained in several studies supports the hypothesis that implants may be successfully osseointegrated when placed immediately after extraction of teeth presenting endodontic and periodontal lesions, provided that appropriate clinical procedures are performed before the implant surgical procedure such as meticulous cleaning, socket curettage/debridement, and chlorhexidine 0.12% rinse. However, more randomized controlled clinical trials with a longer follow-up are required to confirm this procedure as a safe treatment. Moreover, the outcome measures were not related to the type of infection; the classification of infection was often vague and varied among the studies. The benefits of antibiotic solution irrigation and systemic antibiotic administration in such conditions are not yet proved and remain unclear.

[< Back](#)

1 Adell R, Lekholm U, Rockler B, Brånemark PI. A 15-year study of osseointegrated implants in the treatment of the edentulous jaw. *Int J Oral Surg* 1981; **10**: 387-416.

| [CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#) |

2 Naves Mde M, Horbylon BZ, Gomes Cde F, Menezes HH, Bataglion C, Magalhães D. Immediate implants placed into infected sockets: a case report with 3-year follow-up. *Braz Dent J* 2009; **20**: 254-258.

| [PubMed](#) | [Google Scholar](#) |

3 Schropp L, Wenzel A, Kostopoulos L, Karring T. Bone healing and soft tissue contour changes following single-tooth extraction: a clinical and radiographic 12-month prospective study. *Int J Periodontics Restorative Dent* 2003; **23**: 313-323.

| [PubMed](#) | [Web of Science®](#) | [Google Scholar](#) |

4 Schulte W, Heimke G. Das Tübinger Sofort-Implantat. *Quintessenz* 1976; **27**: 17-23.

| [CAS](#) | [PubMed](#) | [Google Scholar](#) |

5 Anneroth G, Hedstrom KG, Kjellman O, Kondell PA, Nordenram A. Endosseus titanium implants in extraction sockets. An experimental study in monkeys. *Int J Oral Surg* 1985; **14**: 50-54.

| [CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#) |

6 Lazzara RJ. Immediate implant placement into extraction sites: surgical and restorative advantages. *Int J Periodontics Restorative Dent* 1989; **9**: 333-343.

| [Google Scholar](#) |

[< Back](#)

Dent Relat Res 2005; **7** (Suppl. 1): S28-S35.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

8 Casap N, Zeltser C, Wexler A, Tarazi E, Zeltser R. Immediate placement of dental implants into debrided infected dentoalveolar sockets. *J Oral Maxillofac Surg* 2007; **65**: 384-392.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

9 Lindeboom JA, Tjiook Y, Kroon FH. Immediate placement of implants in periapical infected sites: a prospective randomized study in 50 patients. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2006; **101**: 705-710.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

10 Werbitt MJ, Goldberg PV. The immediate implant: bone preservation and bone regeneration. *Int J Periodontics Restorative Dent* 1992; **12**: 206-217.

[CAS](#) [PubMed](#) [Google Scholar](#)

11 Rosenquist B, Grenthe B. Immediate placement of implants into extraction sockets: implant survival. *Int J Oral Maxillofac Implants* 1996; **11**: 205-209.

[CAS](#) [PubMed](#) [Google Scholar](#)

12 Quirynen M, Gijbels F, Jacobs R. An infected jawbone site compromising successful osseointegration. *Periodontol 2000* 2003; **33**: 129-144.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

13 Becker W, Becker BE. Guided tissue regeneration for implants placed into extraction sockets and for implant dehiscences: surgical techniques and case report. *Int*

[< Back](#)[CAS](#) | [PubMed](#) | [Google Scholar](#)

14 Barzilay I. Immediate implants: their current status. *Int J Prosthodont* 1993; **6**: 169-175.

[CAS](#) | [PubMed](#) | [Google Scholar](#)

15 Polizzi G, Grunder U, Goené R, et al. Immediate and delayed implant placement into extraction sockets: a 5-year report. *Clin Implant Dent Relat Res* 2000; **2**: 93-99.

[CAS](#) | [PubMed](#) | [Google Scholar](#)

16 Ayangco L, Sheridan PJ. Development and treatment of retrograde peri-implantitis involving a site with a history of failed endodontic and apicoectomy procedures: a series of reports. *Int J Oral Maxillofac Implants* 2001; **16**: 412-417.

[CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

17 Alsaadi G, Quirynen M, Komarek A, van Steenberghe D. Impact of local and systemic factors on the incidence of oral implant failures, up to abutment connection. *J Clin Periodontol* 2007; **34**: 610-617.

[CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

18 Quirynen M, Vogels R, Alsaadi G, Naert I, Jacobs R, van Steenberghe D. Predisposing conditions for retrograde peri-implantitis, and treatment suggestions. *Clin Oral Implants Res* 2005; **16**: 599-608.

[CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

19 Evian CI, Emling R, Rosenberg ES, et al. Retrospective analysis of implant survival and the influence of periodontal disease and immediate placement on long-term results. *Int J Oral Maxillofac Implants* 2004; **19**: 393-398.

[< Back](#)

20 Wagenberg B, Froum SJ. A retrospective study of 1925 consecutively placed immediate implants from 1988 to 2004. *Int J Oral Maxillofac Implants* 2006; **21**: 71-80.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

21 Horwitz J, Zuabi O, Machtei E. Radiographic changes around immediately restored dental implants in periodontally susceptible patients: 1-year results. *Int J Oral Maxillofac Implants* 2008; **23**: 531-538.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

22 Deng F, Zhang H, Zhang H, Shao H, He Q, Zhang P. A comparison of clinical outcomes for implants placed in fresh extraction sockets versus healed sites in periodontally compromised patients: a 1-year follow-up report. *Int J Oral Maxillofac Implants* 2010; **25**: 1036-1040.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

23 Waasdorp JA, Evian CI, Mandracchia M. Immediate placement of implants into infected sites: a systematic review of the literature. *J Periodontol* 2010; **81**: 801-808.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

24 Moher D, Liberati A, Tetzlaff J, Altman DG. The Prisma Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009; **6**: e1000097.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

25 Novaes AB Jr, Novaes AB. Immediate implants placed into infected sites: a clinical report. *Int J Oral Maxillofac Implants* 1995; **10**: 609-613.

[PubMed](#) [Google Scholar](#)

[< Back](#)

269-277.

[CAS](#) | [PubMed](#) | [Google Scholar](#)

27 Balaji A, Nesaline JP, Mohamed JB, Chandrasekaran SC. Placement of endosseous implant in infected alveolar socket with large fenestration defect: a comparative case report. *J Indian Soc Periodontol* 2010; **14**: 270-274.

[PubMed](#) | [Google Scholar](#)

28 Rass MA. Interim endodontic therapy for alveolar socket bone regeneration of infected hopeless teeth prior to implant therapy. *J Oral Implantol* 2010; **36**: 37-59.

[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

29 Viskić J, Milardović S, Katanec D, Vojvodić D, Mehulić K. Immediate implantation in infected tooth sockets. *Coll Antropol* 2011; **35**: 217-221.

[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

30 Pecora G, Andreana S, Covani U, De Leonardis D, Schifferle RE. New directions in surgical endodontics; immediate implantation into an extraction site. *J Endod* 1996; **22**: 135-139.

[CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

31 Novaes AB Jr, Vidigal GM Jr, Novaes AB, Grisi MFM, Polloni S, Rosa A. Immediate implants placed into infected sites. A histomorphometric study in dogs. *Int J Oral Maxillofac Implants* 1998; **13**: 422-427.

[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

[< Back](#)

9: 13-25.

[Google Scholar](#) [↗](#)

33 Tripodakis AP. Sofortimplantation in infizierte und defekte Extraktionsalveolen und kontrollierte Sofortimplantatbelastung. Teil 2: Mehrere benachbarte Zähne. *Implantologie* 2002; **10**: 271-286.

[Google Scholar](#) [↗](#)

34 Rabel A, Köhler SG. Mikrobiologische Untersuchung zur Ermittlung des Erfolges der Sofortimplantation im parodontal geschädigten Gebiss. *Mund Kiefer Gesichtschir* 2006; **10**: 7-13.

[CAS](#) [↗](#) | [PubMed](#) [↗](#) | [Google Scholar](#) [↗](#)

35 Siegenthaler DW, Jung RE, Holderegger C, Roos M, Hammerle CH. Replacement of teeth exhibiting periapical pathology by immediate implants: a prospective, controlled clinical trial. *Clin Oral Implants Res* 2007; **18**: 727-737.

[PubMed](#) [↗](#) | [Web of Science®](#) [↗](#) | [Google Scholar](#) [↗](#)

36 Villa R, Rangert B. Immediate and early function of implants placed in extraction sockets of maxillary infected teeth. A pilot study. *J Prosthet Dent* 2007; **97**: S96-S108.

[PubMed](#) [↗](#) | [Web of Science®](#) [↗](#) | [Google Scholar](#) [↗](#)

37 Del Fabbro M, Boggian C, Taschieri S. Immediate implant placement into fresh extraction sites with chronic periapical pathologic features combined with plasma rich in growth factors: preliminary results of single-cohort study. *J Oral Maxillofac Surg* 2009; **67**: 2476-2484.

[PubMed](#) [↗](#) | [Web of Science®](#) [↗](#) | [Google Scholar](#) [↗](#)

[< Back](#)

Periodontol 2010; **81**: 1140-1146.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

39 Crespi R, Capparè P, Gherlone E. Fresh-socket implants in periapical infected sites in humans. *J Periodontol* 2010; **81**: 378-383.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

40 Bell CL, Diehl D, Bell BM, Bell RE. The immediate placement of dental implants into extraction sites with periapical lesions: a retrospective chart review. *J Oral Maxillofac Surg* 2011; **69**: 1623-1627.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

41 Kusek ER. Immediate implant placement into infected sites: bacterial studies of the Hydroacoustic effects of the YSGG laser. *J Oral Implantol* 2011; **37** (Spec No): 205-211.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

42 Truninger TC, Philipp AO, Siegenthaler DW, Roos M, Hämmerle CH, Jung RE. A prospective, controlled clinical trial evaluating the clinical and radiological outcome after 3 years of immediately placed implants in sockets exhibiting periapical pathology. *Clin Oral Implants Res* 2011; **22**: 20-27.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

43 Fugazzotto P. A retrospective analysis of immediately placed implants in 418 sites exhibiting periapical pathology: results and clinical considerations. *Int J Oral Maxillofac Implants* 2012; **27**: 194-202.

[PubMed](#) [Web of Science®](#) [Google Scholar](#) 

[< Back](#)[PubMed](#) [Web of Science®](#) [Google Scholar](#)

45 Jofre J, Valenzuela D, Quintana P, Asenjo-Lobos C. Protocol for immediate implant replacement of infected teeth. *Implant Dent* 2012; **21**: 287-294.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

46 Jung RE, Zaugg B, Philipp AO, Truninger TC, Siegenthaler DW, Hammerle CH. A prospective, controlled clinical trial evaluating the clinical radiological and aesthetic outcome after 5 years of immediately placed implants in sockets exhibiting periapical pathology. *Clin Oral Implants Res* 2012 (in press).

[Google Scholar](#)

47 Meltzer AM. Immediate implant placement and restoration in infected sites. *Int J Periodontics Restorative Dent* 2012; **32**: e169-e173.

[PubMed](#) [Google Scholar](#)

48 Marconcini S, Barone A, Gelpi F, Briguglio F, Covani U. Immediate implant placement in infected sites: a case series. *J Periodontol* 2013; **84**: 196-202.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

49 Marcaccini AM, Novaes AB Jr, Souza SL, Taba M Jr, Grisi MF. Immediate placement of implants into periodontally infected sites in dogs. Part 2: a fluorescence microscopy study. *Int J Oral Maxillofac Implants* 2003; **18**: 812-819.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

50 Novaes AB Jr, Marcaccini AM, Souza SLS, Taba M Jr, Grisi MFM. Immediate placement of implants into periodontally infected sites in dogs: a histomorphometric

[< Back](#)[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

51 Tehemar S, Hanes P, Sharawy M. Enhancement of osseointegration of implants placed into extraction sockets of healthy and periodontally diseased teeth by using graft material, an ePTFE membrane, or a combination. *Clin Implant Dent Relat Res* 2003; **5**: 193-211.

[CAS](#) | [PubMed](#) | [Google Scholar](#)

52 Novaes AB Jr, Papalexiou V, Grisi MF, Souza SS, Taba M Jr, Kajiwarra JK. Influence of implant microstructure on the osseointegration of immediate implants placed in periodontally infected sites. A histomorphometric study in dogs. *Clin Oral Implants Res* 2004; **15**: 34-43.

[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

53 Papalexiou V, Novaes AB Jr, Grisi MF, Souza SS, Taba M Jr, Kajiwarra JK. Influence of implant microstructure on the dynamics of bone healing around immediate implants placed into periodontally infected sites. A confocal laser scanning microscopic study. *Clin Oral Implants Res* 2004; **15**: 44-53.

[CAS](#) | [PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

54 Chang SW, Shin SY, Hong JR, et al. Immediate implant placement into infected and noninfected extraction sockets: a pilot study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2009; **107**: 197-203.

[PubMed](#) | [Web of Science®](#) | [Google Scholar](#)

55 Brisman DL, Brisman AS, Moses MS. Implant failures associated with asymptomatic endodontically treated teeth. *J Am Dent Assoc* 2001; **132**: 191-195.

[CAS](#) | [PubMed](#) | [Google Scholar](#)

[< Back](#)[PubMed](#) [Web of Science®](#) [Google Scholar](#)

57 Sunde PT, Tronstad L, Eribe ER, Lind PO, Olsen I. Assessment of periradicular microbiota by DNA-DNA hybridization. *Endod Dent Traumatol* 2000; **16**: 191-196.

[CAS](#) [PubMed](#) [Web of Science®](#) [Google Scholar](#)

58 Nelson S, Thomas G. Bacterial persistence in dentoalveolar bone following extraction: a microbiological study and implications for dental implant treatment. *Clin Implant Dent Relat Res* 2010; **12**: 306-314.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

59 Kassolis JD, Scheper M, Jham B, Reynolds MA. Histopathologic findings in bone from edentulous alveolar ridges: a role in osteonecrosis of the jaws? *Bone* 2010; **47**: 127-130.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

60 Peters LB, Wesselink PR, van Winkelhoff AJ. Combinations of bacterial species in endodontic infections. *Int Endod J* 2002; **35**: 698-702.

[CAS](#) [PubMed](#) [Web of Science®](#) [Google Scholar](#)

61 Drago CJ, Lazzara RJ. Immediate provisional restoration of Osseotite implants: a clinical report of 18- month results. *Int J Oral Maxillofac Implants* 2004; **19**: 534-541.

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

62 Dent CD, Olson JW, Farish SE, et al. The influence of preoperative antibiotics on success of endosseous implants up to and including stage II surgery: a study of 2,641 implants. *J Oral Maxillofac Surg* 1997; **55** (Suppl 5): 19-24.

[< Back](#)

63 Esposito M, Grusovin MG, Talati M, Coulthard P, Oliver R, Worthington HV. Interventions for replacing missing teeth: antibiotics at dental implant placement to prevent complications. *Cochrane Database Syst Rev* 2008; **3**:CD004152.

[Google Scholar](#) [↗](#)

64 Sae-Lim V, Metzger Z, Trope M. Local dexamethasone improves periodontal healing of replanted dogs' teeth. *Endod Dent Tramadol* 1998; **14**: 232-236.

[CAS](#) [↗](#) | [PubMed](#) [↗](#) | [Web of Science®](#) [↗](#) | [Google Scholar](#) [↗](#)

65 Lambert PM, Morris HF, Ochi S. The influence of 0.12% chlorhexidine digluconate rinses on the incidence of infectious complications and implant success. *J Oral Maxillofac Surg* 1997; **55** (Suppl 5): 25-30.

[CAS](#) [↗](#) | [PubMed](#) [↗](#) | [Web of Science®](#) [↗](#) | [Google Scholar](#) [↗](#)

66 Kido H, Schulz EE, Kumar A, Lozada J, Saha S. Implant diameter and bone density: effect on initial stability and pull out resistance. *J Oral Implantol* 1997; **23**: 163-169.

[CAS](#) [↗](#) | [PubMed](#) [↗](#) | [Google Scholar](#) [↗](#)

Citing Literature



Related jobs

Medical Lab Scientist- Per Diem

- Competitive Salary
- Nashua, United States

[< Back](#)**MEDICAL LABORATORY SCIENTIST - LA**

- Competitive Salary
- Rhinelander, Wisconsin, United States
- Aspirus Health

Ast, Aso or Full Professor

- Competitive Salary
- Gainesville, Florida, United States

[Download PDF](#)**ABOUT WILEY ONLINE LIBRARY**[Privacy Policy](#)[Terms of Use](#)[About Cookies](#)[Manage Cookies](#)[Accessibility](#)[Wiley Research DE&I Statement and Publishing Policies](#)**HELP & SUPPORT**[Contact Us](#)[Training and Support](#)[DMCA & Reporting Piracy](#)[Sitemap](#)**OPPORTUNITIES**[Subscription Agents](#)[Advertisers & Corporate Partners](#)**CONNECT WITH WILEY**[The Wiley Network](#)[Wiley Press Room](#)

< Back

and data mining and training of artificial intelligence technologies or similar technologies.